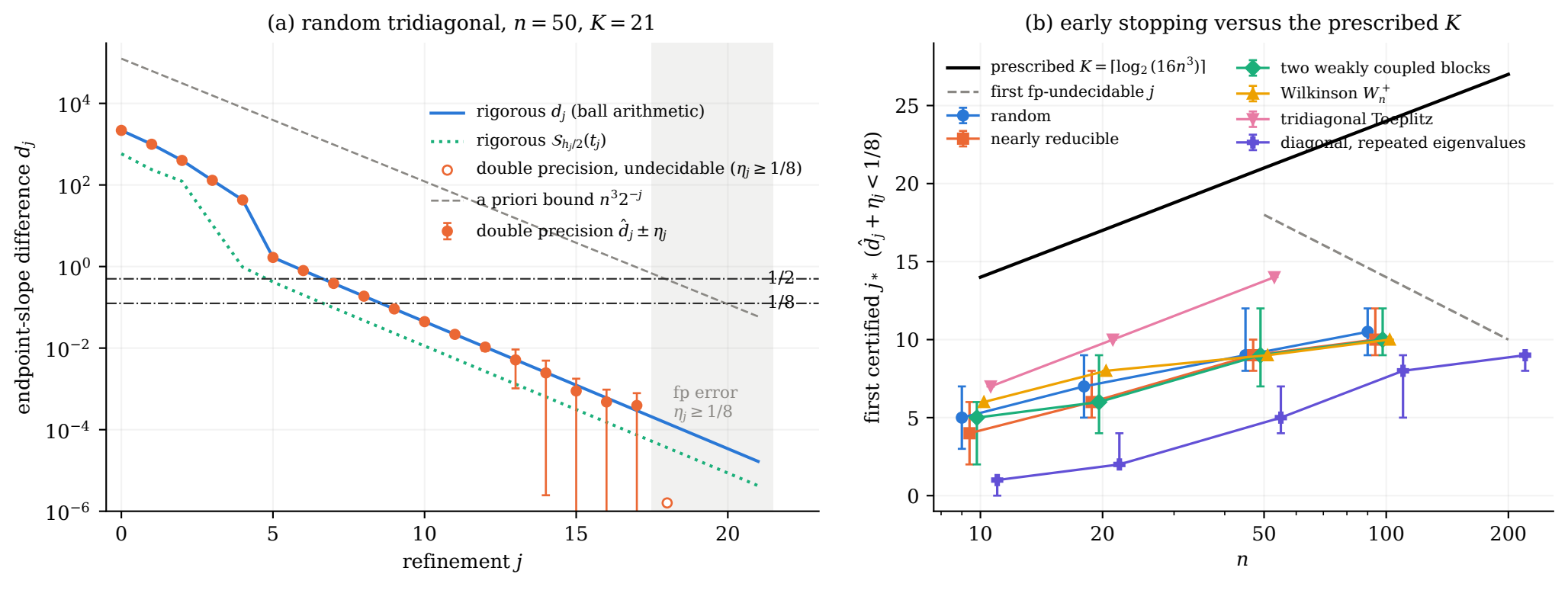
(a) random tridiagonal, $n = 50$, $K = 21$
endpoint-slope difference d_j
refinement j
rigorous d_j (ball arithmetic)
rigorous $s_{h_j/2}(t_j)$
double precision, undecidable ($\eta_j \geq 1/8$)
a priori bound $n^3 2^{-j}$
double precision $\hat{d}_j \pm \eta_j$
1/2
1/8
fp error
$\eta_j \geq 1/8$
(b) early stopping versus the prescribed K
first certified j_* ($\hat{d}_j + \eta_j < 1/8$)
n
prescribed $K = \lceil \log_2(16n^3) \rceil$
first fp-undecidable j
random
nearly reducible
two weakly coupled blocks
Wilkinson W_n^+
tridiagonal Toeplitz
diagonal, repeated eigenvalues